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RESEARCH ARTICLE

Tallgrass prairie restoration: implications of increased atmospheric nitrogen deposition when site preparation minimizes adventive grasses

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Site preparation designed to exhaust the soil seedbank of adventive species can improve the success of tallgrass prairie restoration. Despite these efforts, increased rates of atmospheric nitrogen (N) deposition over the next century could potentially promote the growth of nitrophilic, adventive species in tallgrass restoration projects. We used a field experiment to examine how N addition affected species composition and plant productivity over the first 3 years of a tallgrass prairie restoration that was preceded by the planting of glyphosate-resistant crops and multiple applications of glyphosate to exhaust the pre-existing seedbank. We predicted that N addition would increase the percent cover of adventive plant species not included in the original seeding. Contrary to our prediction, only the cover of native species increased with N addition; native non-leguminous forbs increased substantially, with *Conyza canadensis* (a weedy native species not part of the restoration seed mix) exploiting the combination of high N and bare ground in the first year, and non-leguminous forbs (in particular *Monarda fistulosa*) and native C_3 grasses, all of which were seeded, increasing with N addition by the third year. Native legumes was the only functional group that exhibited lower cover in N addition plots than in control plots. There was no significant response by native C_4 grasses to N addition, and adventive grasses remained mostly absent from the plots. Overall, our results suggest that site pre-treatment with herbicide may continue to be effective in minimizing adventive grasses in restored tallgrass prairie, despite future increases in atmospheric N deposition.

Key words: atmospheric nitrogen deposition, cover, legumes, site preparation, tallgrass prairie restoration

Implications for Practice

- Herbicide pre-treatment to exclude adventive C_3 grasses from tallgrass prairie restoration projects may remain effective despite increased atmospheric N deposition.
- The enhanced invasion of prairie restoration sites by weedy species in high N plots was only temporary, diminishing after 1 year.
- Nitrogen deposition may have adverse effects on the persistence of legumes in restored tallgrass prairie, whereas other native forbs and native C_3 bunchgrasses may be favored.

Introduction

Agricultural intensification and the release of reactive nitrogen (N) from fossil fuel combustion have increased emissions of reactive N into the atmosphere over the last century, and much of this reactive N is subsequently deposited across the landscape in the form of NH_3 and NO_3^- compounds in dust or precipitation (Galloway et al. 2004). Plant growth in most terrestrial ecosystems is limited by N availability (Lebauer & Treseder 2008), and in field experiments examining the combined effects of global change factors, N addition is consistently among the most influential factors affecting plant productivity and community composition (Torok et al. 2000; Dukes et al. 2005; Miles & Knops 2009). Atmospheric N deposition has thus emerged as one of the most critical threats to native species diversity (Payne et al. 2013), especially in areas that have experienced elevated deposition rates for several decades (Phoenix et al. 2006).

In the context of plant community restoration and global environmental change, the question has arisen as to whether plant communities are best restored to their historical species composition or whether attempts should be made to develop restoration goals under anticipated future environmental conditions (Temperton et al. 2004). Grasslands are systems where plant productivity and species composition have already been influenced considerably by atmospheric N deposition in some regions (Stevens et al. 2004, 2010) and future grassland restoration efforts likewise may be vulnerable to increased N deposition. By recognizing the key features of plant establishment responses to increased N availability, property managers will

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be able to maximize the effectiveness of grassland restoration strategies (Torok et al. 2000).

Tallgrass prairie is one of the most severely degraded plant community types in North America (Packard & Mutel 1997); 162 million ha of tallgrass prairie were estimated to be present on this continent in 1830, but this area has declined by 82–99% (Samson & Knopf 1994). Restoration of these communities has increased in recent decades, particularly in areas of marginal agricultural potential, such as sand plains (Delaney et al. 2000). Tallgrass prairies are typically characterized by low N availability (Schramm 1990; Seastedt et al. 1991). The C_4 grasses that dominate these systems have high N use efficiency, which can give them a competitive advantage over C_3 species when N is strongly limited (Wilson & Tilman 1991). However, increased N availability can significantly alter the species composition of intact tallgrass praire (Wedin & Tilman 1990; Torok et al. 2000), often in favor of fast-growing adventive species (Miles & Knops 2009). N is likewise a key soil resource that regulates productivity and diversity in newly established tallgrass prairie (Baer et al. 2003). The first few years of a tallgrass restoration project may be crucial for determining its success, because once tallgrass prairies mature and the species become well established, the ability of fast-growing adventive species to successfully colonize the area may be reduced (Gartshore 2011). Site preparation measures, such as multiple applications of herbicide with low residual activity, can improve the success of tallgrass prairie restoration projects by exhausting the soil seedbank of adventive species (Williams 2010). Nevertheless, the continued effectiveness of these site preparation measures in the presence of increased atmospheric N deposition is unknown.

The increased success of non-tallgrass species relative to tallgrass species could ultimately decrease the biodiversity of these systems (Clark & Tilman 2008; Miles & Knops 2009). Many adventive C_3 grasses, for example, can reproduce vegetatively by extending aboveground stolons or belowground roots and rhizomes that interweave to form a dense mat, making it difficult for other species to compete for space (McGregor et al. 1991). Likewise, much of the root growth in these C_3 turf-forming grasses is horizontal, which makes it difficult for other species to compete for nutrients and moisture (Barbour et al. 1999). In contrast, the C_4 bunchgrasses characteristic of tallgrass prairie form clumps with a high vertical component of root growth, which allows other species to grow in between these patches (McGregor et al. 1991). Therefore, it is possible that the negative effects of increased N availability on the diversity of prairie species may be minimized if C_3 species are not present.

We explored how increased atmospheric N deposition may influence the establishment of plants in restored tallgrass prairie by performing an N addition experiment. In contrast to previous studies, we were particularly interested in examining these effects at a site where the presence of turf-forming C_3 grasses was limited by the planting of glyphosate-resistant crops and treatment with multiple applications of glyphosate to exhaust the pre-existing soil seedbank. We applied N at rates consistent with both moderate ($2\,g\,N\,m^{-2}\,yr^{-1}$) and high ($6\,g\,N\,m^{-2}\,yr^{-1}$) rates of atmospheric N deposition for the region, and recorded changes in plant species composition, cover, and biomass over the first three growing seasons following the initial restoration, and compared these plots with those that received no added N. We predicted that N addition would disproportionately favor the increased cover and biomass of species that were not part of the initial tallgrass restoration seeding, and in particular adventive species. We also predicted that N addition would decrease plant species diversity.

Methods

Study Site

Our experiment was conducted in a 20.9 ha tallgrass prairie site (42.687078, −80.466565) in Norfolk County, Ontario, Canada, that was restored by the Nature Conservancy of Canada in the spring of 2010. The site was located on a sand plain (mean soil pH ± SE [standard error]: 6.77 ± 0.07, $n = 24$), and was previously used as a tobacco farm until 2003. The site was planted with rotations of glyphosate-resistant corn and soybean in the interim, and treated again with glyphosate herbicide prior to the sowing of the tallgrass species in 2010. Adventive grasses that established within 10 m of the periphery of the site were also controlled with glyphosate over the course of the study. For the restoration, seeds (collected from local genotypes) were sown from a mix of native, tallgrass prairie species (Table 1). The cost of the restoration was approximately $1000 USD per hectare.

Experimental Design

In June 2010, we established N addition plots arranged in eight experimental blocks at 25-m intervals along a transect. Each block consisted of a set of three 2×2-m plots spaced at least 1.5 m apart, and N treatments (0, 2, or $6\,g\,N\,m^{-2}\,yr^{-1}$) were randomly assigned to the plots within each block, with N added every year in early May in the form of slow release ammonium nitrate Osmocote pellets. These rates were chosen to represent the range of the current deposition rate, and moderate, and high projections for N deposition rates in this region by the year 2050 (Galloway et al. 2004). We included a 10 cm buffer zone of N addition around each plot to minimize edge effects.

Sample Collection

We estimated plant cover on a per species basis (Domin-Krajina cover-abundance scale; Mueller-Dombois & Ellenberg 1974) in August and September in 2011, and at 1-month intervals from May to October in both 2012 and 2013, from 1×1 m subplots of each plot. We also collected the standing aboveground biomass of all species from 25×25 cm subplots in mid-July and the beginning of September in 2012. After sorting aboveground material to the species level, we dried it for at least 48 h at 65°C prior to weighing. We also collected two 2 cm diameter, 15 cm deep soil cores from each plot in August 2011, September 2012, and June 2013 to measure root biomass. We

Table 1. Percentage of total seed mass sown, and annual maximum of mean cover for (a) native species and (b) adventive species ± SE (standard error), $n = 8$.

		Maximum cover—2011						*Maximum cover—2012*						*Maximum cover—2013*					
		0 g m^{-2}		*2 g m^{-2}*		*6 g m^{-2}*		*0 g m^{-2}*		*2 g m^{-2}*		*6 g m^{-2}*		*0 g m^{-2}*		*2 g m^{-2}*		*6 g m^{-2}*	
	% seed Mass	*Mean*	*SE*	*Mean*	*SE*	*Mean*	*SE*	*Mean*	*SE*	*Mean*	*SE*	*Mean*	*SE*	*Mean*	*SE*	*Mean*	*SE*	*Mean*	*SE*
(a) Native species																			
(i) Grasses (seeded)																			
Andropogon gerardii	13.1	2.9	1.3	1.2	0.8	4.6	3.0	6.1	2.6	3.9	3.5	19.0	5.8	13.9	5.3	5.1	2.7	10.8	3.3
Bromus kalmii	3.3	1.0	0.5	1.3	0.8	1.7	1.0	3.0	2.2	5.3	3.5	5.9	5.0	5.0	1.0	11.7	3.5	13.1	4.1
Elymus trachycalus	6.6	2.8	1.0	2.9	2.2	2.1	0.9	0.3	0.3	1.2	0.9	8.7	3.9	1.9	0.8	8.1	3.5	9.9	2.2
Schizachyrium scoparium	21.9	14.9	4.1	9.3	4.3	9.1	3.1	21.0	4.7	13.4	5.1	22.3	6.9	21.6	3.7	22.5	5.3	23.2	4.4
Sorghastrum nutans	13.1	11.0	3.9	5.6	2.4	7.6	2.8	20.4	8.3	11.3	4.2	16.2	8.2	15.8	4.0	9.6	5.6	10.6	3.1
(ii) Non-leguminous forbs (seeded)																			
Artemisia campestris	1.3	7.6	4.4	0.8	0.5	1.8	1.2	3.1	2.2	1.8	1.1	0.3	0.3	1.0	0.9	0.8	0.4	2.1	2.1
Asclepias syriaca	0.2	—	—	0.1	0.1	0.4	0.4	0.1	0.1	0.1	0.1	2.1	2.1	—	—	—	—	—	—
Asclepias tuberosa	4.4	0.2	0.1	1.4	1.3	0.4	0.2	—	—	0.1	0.1	0.1	0.1	—	—	—	—	0.3	0.3
Euthamia graminifolia	0.2	—	—	—	—	—	—	2.2	2.1	—	—	—	—	0.7	0.4	0.1	0.1	0.9	0.9
Gnaphalium obtusifolium	0.2	—	—	—	—	—	—	—	—	—	—	0.1	0.1	—	—	—	—	—	—
Liatris cylandracea	4.4	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Monarda fistulosa	1.3	1.1	0.8	1.1	0.8	1.8	1.0	4.2	2.1	3.7	2.1	9.2	3.3	12.0	3.6	10.2	4.8	35.4	10.5
Oenthera biennis	2.2	0.8	0.5	0.8	0.5	0.1	0.1	0.9	0.9	0.7	0.4	0.9	0.9	—	—	—	—	—	—
Pycnanthemum virginianum	0.9	2.2	2.2	3.5	2.2	0.9	0.7	—	—	3	2.2	2.1	2.1	0.1	0.1	8.8	7.7	3.6	3.6
Rudbeckia hirta	1.3	10.0	5.4	9.6	3.0	20.1	4.8	15.3	3.6	16.8	4.6	22.8	6.0	14.6	4.6	19.7	5.0	17.6	3.5
Solidago juncea	0.9	3.6	2.0	2.8	2.2	0.9	0.9	5.8	3.9	13.0	5.9	4.3	2.8	6.9	3.7	8.7	3.9	6.3	2.5
Solidago nemoralis	0.9	3.1	1.4	12.7	5.7	8.3	3.7	—	—	—	—	—	—	—	—	—	—	—	—
Symphypotrichum ericoides	0.2	0.9	0.6	2.9	1.0	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Symphyotrichum laeve	1.3	0.4	0.4	0.1	0.1	2.3	2.2	9.6	3.7	10.4	3.4	8.1	2.8	4.9	2.7	11.3	3.2	6.3	2.5
Symphyotrichum oolentangiense	4.4	4.5	1.9	3.4	1.3	5.1	1.6	—	—	—	—	—	—	3.7	2.1	3.4	2.5	5.5	2.6
Symphyotrichum pilosus	1.1	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Symphyotrichum urophyllum	0.7	0.7	0.4	0.3	0.1	2.1	0.9	6.1	2.6	8.6	3.2	11.7	5.6	1.5	0.9	6.0	3.8	1.5	0.9
(iii) Non-leguminous forbs (not seeded)																			
Ambrosia artemisiifolia	—	2.0	1.0	0.9	0.7	1.0	0.5	0.1	0.1	2.2	2.1	0.2	0.1	0.1	0.1	—	—	0.4	0.3
Arabidopsis lyrata	—	—	—	—	—	—	—	1.3	0.9	9.1	2.5	14.4	3.9	5.7	1.9	5.7	2.0	6.6	1.8
Conyza canadensis	—	15.1	4.8	31.0	8.3	39.2	6.2	5.3	2.7	12.9	7.3	7.8	4.9	4.5	2.1	4.3	2.1	4.6	2.7
Erigeron annuus	—	—	—	—	—	—	—	0.3	0.3	—	—	0.3	0.3	0.9	0.9	—	—	3.6	3.6
Gnaphalium viscosum	0.02	0.1	0.1	—	—	0.1	0.1	—	—	—	—	—	—	—	—	—	—	—	—
Oxalis dilenii	—	—	—	0.3	0.2	1.6	1.0	0.1	0.1	1.0	0.9	4.8	2.1	—	—	2.3	2.1	0.9	0.9
Symphyotrichum lanceolatum	—	0.4	0.4	6.2	3.9	2.6	2.2	—	—	—	—	—	—	—	—	—	—	—	—
(iv) Leguminous forbs (seeded)																			
Desmodium canadense	2.2	6.0	4.3	2.0	0.9	1.5	0.6	2.7	1.3	0.3	0.3	0.3	0.3	9.9	4.0	0.1	0.1	2.2	2.1
Desmodium paniculatum	2.2	3.9	2.4	3.6	2.2	1.0	0.5	—	—	—	—	—	—	—	—	—	—	—	—
Lespedeza capitata	4.4	0.9	0.5	0.7	0.4	1.3	0.5	4.8	2.7	2.2	2.1	3.1	2.2	4.9	3.4	1.6	0.9	1.1	0.4
Lupinus perennis	2.2	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
(v) Horsetail (not seeded)																			
Equisetumarvense	—	4.1	1.7	0.5	0.4	0.6	0.5	2.1	1.1	0.4	0.3	1.2	0.9	6.9	3.7	2.8	2.1	1.3	0.9
(b) Adventive species (not seeded)																			
(i) Non-leguminous forbs																			
Anthemis cotula	—	0.5	0.4	2.7	2.1	3.4	2.1	0.9	0.9	1.9	1.1	2.2	1.1	—	—	—	—	—	—
Cerastium fontanum	—	1.5	0.6	3.8	1.7	2.4	1.9	—	—	—	—	—	—	—	—	—	—	—	—
Crepis capillaris	—	—	—	—	—	—	—	—	—	—	—	2.1	2.1	—	—	—	—	—	—
Holosteum umbellatum	—	—	—	—	—	—	—	32.9	12.3	30.4	12.6	18.7	10.7	4.7	2.0	3.3	2.1	4.8	3.5
Taraxacum officinale	—	4.0	3.0	4.3	2.4	6.0	2.4	4.4	1.3	11.1	3.6	14.7	2.6	3.0	2.2	6.6	2.5	5.0	1.0
Veronica arvensis	—	—	—	—	—	—	—	7.8	2.3	5.6	2.0	3.9	2.2	7.7	1.4	11.6	3.0	5.8	1.8
(ii) Leguminous forbs																			
Medicago lupulina	—	0.4	0.4	0.1	0.1	—	—	5.1	5.1	2.1	2.1	2.1	2.1	1.8	1.1	2.5	2.1	0.9	0.9

Species with <2% of initial seed mass, not present in plots: *Doellingeria umbellata*, *Hedyotis longifolia*, *Helianthemum bicknelli*, *Helianthus divaricatus*, *Lechea villosa*, *Lespedeza hirta*, *Lespedeza intermedia*, *Mainanthemum stellatum*, *Penstemon digitalis*, *Scirpus cyperinus*, *Silene antirrhina*, *Sisyrhincium montanum*, *Specularia perfoliata*, *Sporobolus cryptandrus*.

Species not seeded, cover <1%: *Arenaria serpyllifolia*, *Cerastium arvense*, *Chenopodium album*, *Crepistectorum*, *Lactuca spp.*, *Setaria viridis*, *Solidago canadensis*, *Symphyotrichum novae-angliae*, *Trifolium hybridum*, *Trifolium repens*.

washed the soil from the roots using sieves and reverse osmosis water, then dried them at 65°C for at least 48 h prior to weighing.

Data Analyses

For each growing season, we determined the maximum monthly percent cover value for each species in each plot to account for differences in the seasonality of peak cover among species. We then summed these values for each plot to obtain total percent cover estimates for native versus adventive species separated into each of the main plant functional groups (C_3 grasses, C_4 grasses, non-leguminous forbs [seeded and not seeded analyzed separately], leguminous forbs, and horsetails). The same approach was taken for analyzing the aboveground biomass data. For data collected over multiple years, we assessed the treatment effects on all response variables individually for each combination of native vs. adventive status and functional group using randomized block repeated measures analysis of variances (ANOVAs) that included N addition as the between plot factor, and date and the interaction between date and N addition as within-plot factors (plot was included as a random factor nested in N addition). We ran the analyses using the restricted maximum likelihood (REML) method of the Fit Model platform in JMP 4.0 (SAS Institute, Cary, NC, U.S.A.). For the biomass data, which were collected for a single year, we used one-way ANOVA. All cover and biomass data were square-root transformed to satisfy the ANOVA assumptions of normality and homogeneity of variances. Community diversity was calculated based on the cover estimates using the Shannon diversity index (H'), and H' and species richness were used to calculate evenness (J'). Treatment effects on species diversity, richness, and evenness were analyzed using the same ANOVA models described for the cover data above.

Results

All three plant growing seasons (March-October) featured mean air temperatures ranging from near normal to above normal (16.9, 16.1, and 17.6°C for 2011, 2012, and 2013, respectively, compared to the 1971–2000 climate normal of 16.3°C – Environment Canada, National Climate Data, and Information Archive). For mean monthly precipitation, the second growing season was notably dry (100, 54, and 78 mm for 2011, 2012, and 2013, respectively, compared to the 1971–2000 climate normal of 87 mm – Environment Canada, National Climate Data, and Information Archive), and it also followed a winter that was 2.8°C warmer than normal and with half the normal amount of precipitation.

Total percent cover (i.e. percent cover values summed over all species) increased with N addition ($P = 0.01$), and it was significantly higher in the second and third years than in the first year ($P < 0.001$) (Fig. 1). Much of the N addition response was explained by the significant increase in non-leguminous native forbs with N addition ($P = 0.011$ and $P = 0.004$ for seeded and non-seeded species, respectively; Fig. 2). In the first year following the restoration, the N effect on non-leguminous forbs

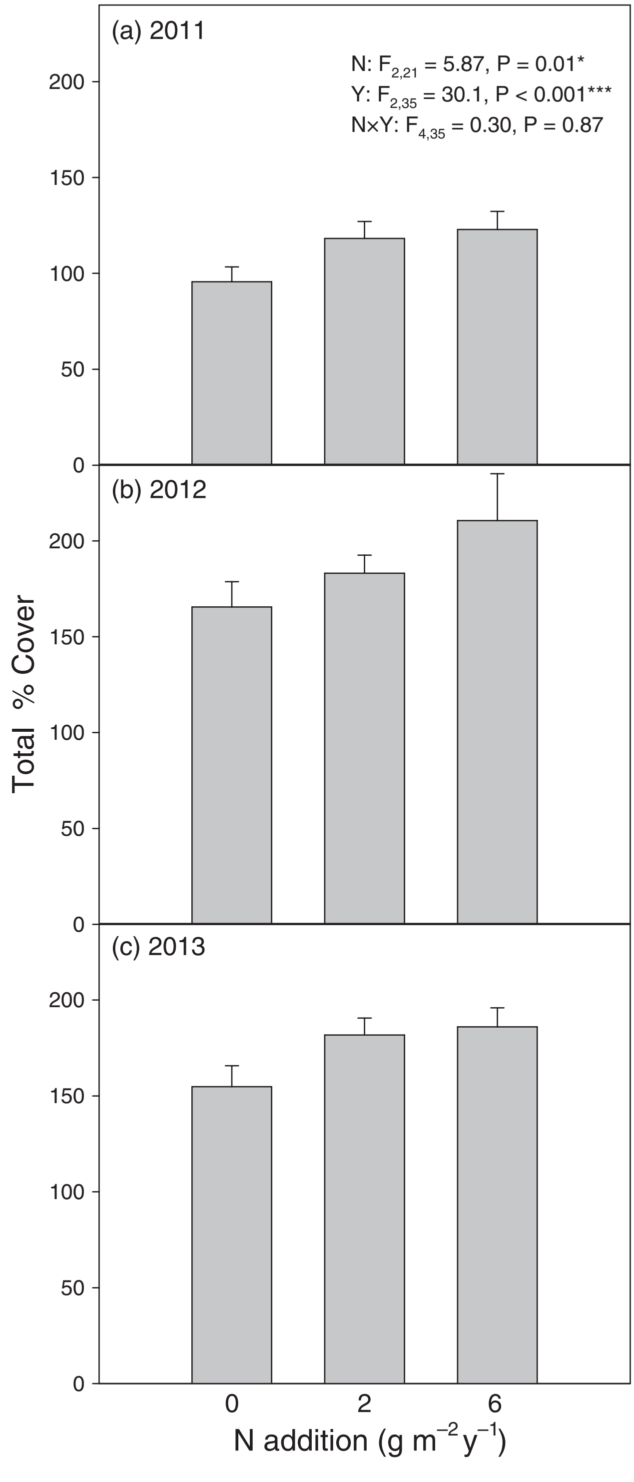

Figure 1. Mean total percent plant cover (the sum of percent cover values for all species) in response to N addition in (a) 2011, (b) 2012, and (c) 2013. Error bars represent standard error ($n = 8$). Repeated measures ANOVA results are displayed (N – N addition, Y – Year, N × Y – Interaction, $*P < 0.05$, $**P < 0.01$, $***P < 0.001$).

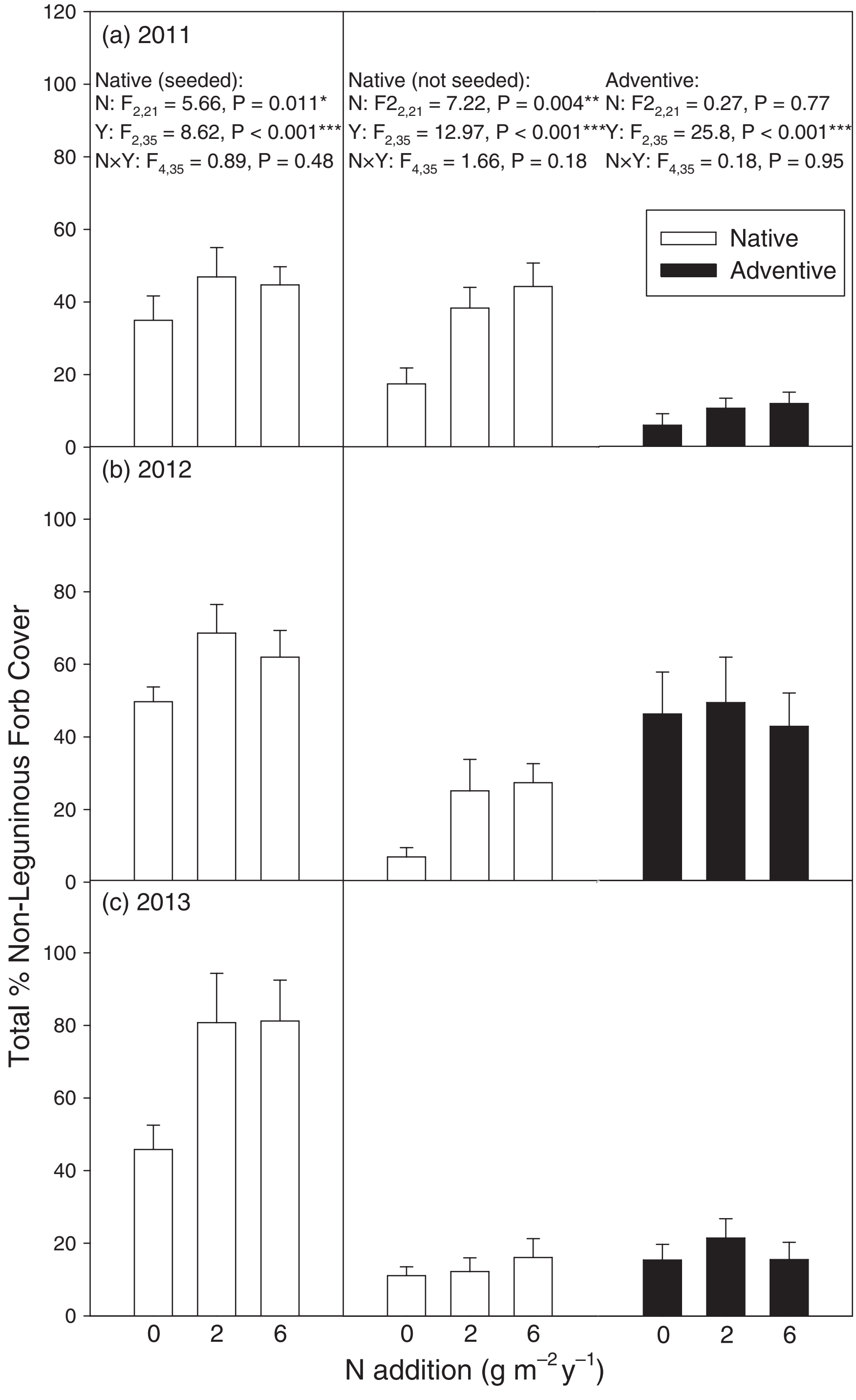

Figure 2. Mean total percent cover of native non-leguminous forbs (seeded and not seeded − both indicated by white bars) and adventive non-leguminous forbs (black bars) in response to N addition in (a) 2011, (b) 2012, and (c) 2013. Error bars represent standard error ($n = 8$). Repeated measures ANOVA results are displayed (N – N addition, Y – Year, N × Y – Interaction, $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$).

that were not seeded was primarily driven by increases in the cover of *Conyza canadensis* (mean cover of 15, 31, and 39% for the 0, 2, and $6\,g\,N\,m^{-2}\,yr^{-1}$ treatments, respectively; $P = 0.010$), a native weedy species that was not included in the restoration seed mix. However, *C. canadensis* cover declined in the subsequent year, whereupon *Arabidopsis lyrata* was responsible for much of the N addition effect on non-seeded, non-leguminous forbs (Table 1). For the non-leguminous forbs that were seeded, although there were no significant differences between responses to moderate and high N addition at

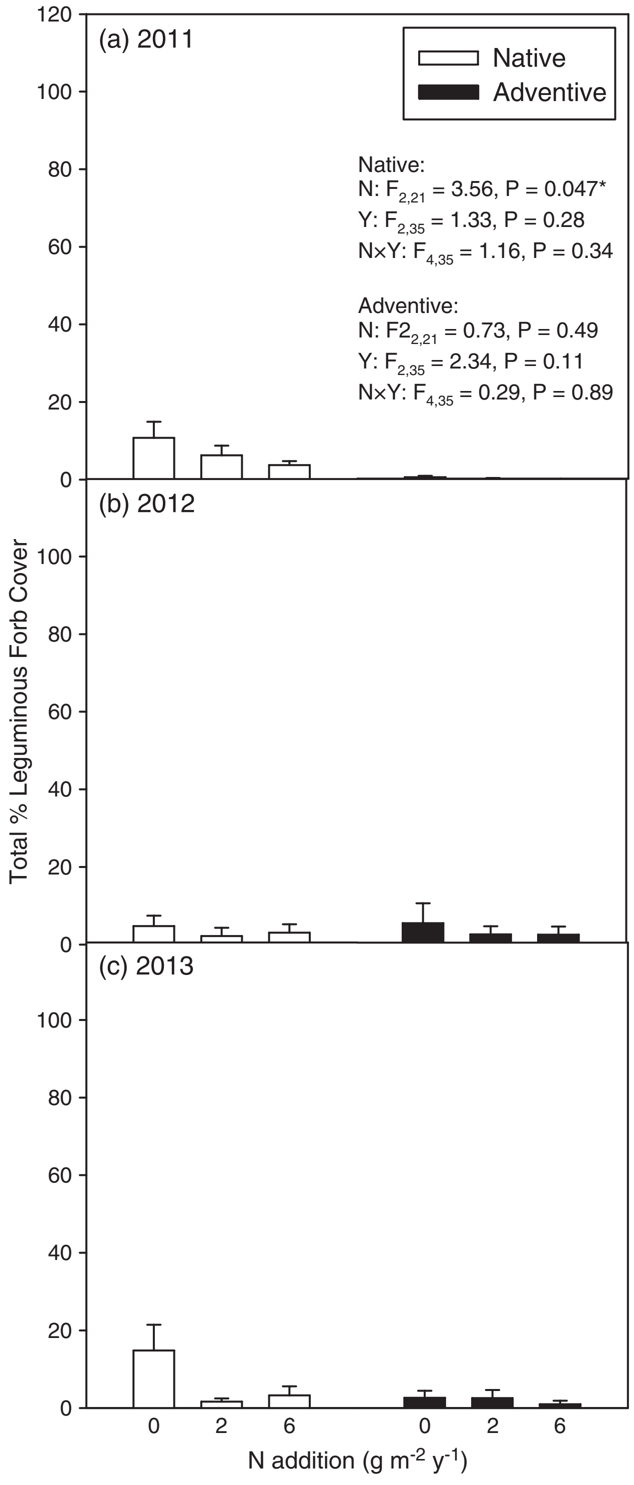

Figure 3. Mean total percent cover of native leguminous forbs (white bars) and adventive leguminous forbs (black bars) in response to N addition in (a) 2011, (b) 2012, and (c) 2013. Error bars represent standard error ($n = 8$). Repeated measures ANOVA results are displayed (N – N addition, Y – Year, N × Y – Interaction, *$P < 0.05$, **$P < 0.01$, ***$P < 0.001$).

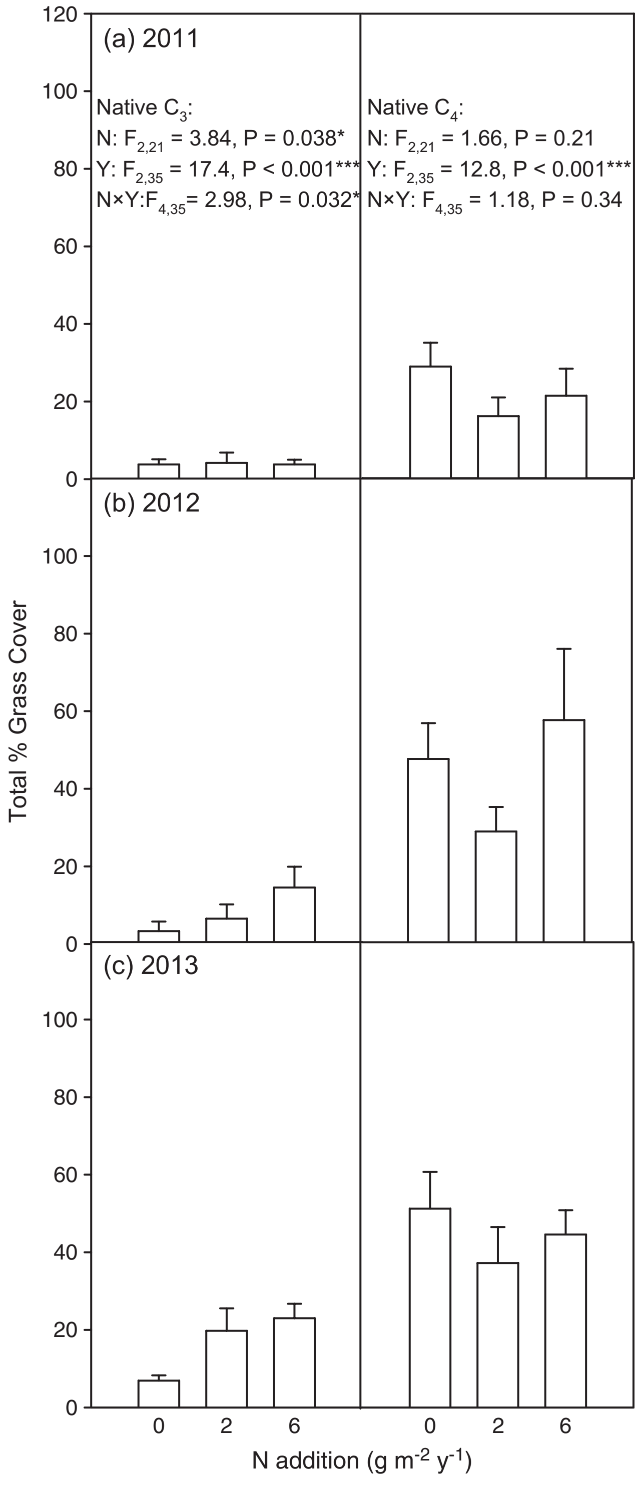

Figure 4. Mean total percent cover of native C_3 (left panels) and native C_4 (right panels) grasses in response to N addition in (a) 2011, (b) 2012, and (c) 2013. Error bars represent standard error ($n = 8$). Repeated measures ANOVA results are displayed (N – N addition, Y – Year, N × Y – Interaction, *$P < 0.05$, **$P < 0.01$, ***$P < 0.001$).

the functional group level in any of the years, by the third year the N addition response in the moderate N addition plots was explained by increases in a consortium of species, mainly *Pycnanthemum virginianum*, *Rudbeckia hirta*, *Solidago juncea*, *Symphyotrichum laeve*, and *Symphyotrichum urophyllum* (Table 1). In contrast, the latter species were not highly abundant in the high N addition plots, and increases in the cover of seeded non-leguminous forbs were instead driven primarily by increases in *Monarda fistulosa* (Table 1). Unlike the non-leguminous forbs, native leguminous forbs were significantly lower in the N addition plots than the control plots ($P = 0.047$; Fig. 3), with this result occurring for the two most abundant legume species, *Desmodium canadense* and *Lespedeza capitata* (Table 1). Increases in native grass cover in years two and three relative to year one were primarily responsible for the increase in total percent cover over time, with variation among species in grass cover in the N-free plots mostly consistent by year three with the relative proportions of seeds included in the initial sowing (Table 1); there was a significant increase in total native C_3 grass cover in response to N addition in the second and third years (N × year interaction: $P = 0.032$), but there were no significant N addition effects on total native C_4 grass cover (Fig. 4). Adventive grasses were almost entirely absent from the plots over the course of the study.

There were no significant effects of N addition on total aboveground biomass ($P = 0.93$; means ± SE: 973 ± 210, 1088 ± 212, and 1101 ± 246 g m^{-2} for the 0, 2, and 6 g N m^{-2} treatments, respectively; $n = 8$), or the aboveground biomass of native and adventive species ($P = 0.93$ and 0.32, respectively). Root biomass was also not significantly affected by N addition ($P = 0.74$; overall means pooled over all treatments ± SE: 531 ± 72, 977 ± 72, and 734 ± 83 g m^{-2} for 2011, 2012, and 2013, respectively; $n = 24$). Likewise, there were no significant effects of N addition on the Shannon diversity index ($P = 0.66$), species richness ($P = 0.79$), or species evenness ($P = 0.77$) (means pooled over all treatments ± SE, $n = 24$, for (1) diversity: 2.09 ± 0.06, 2.39 ± 0.03, and 2.28 ± 0.04 for 2011, 2012, and 2013, respectively; (2) species richness: 15.5 ± 0.7, 12.3 ± 0.4, and 15.3 ± 0.4 for 2011, 2012, and 2013, respectively, (3) evenness: 0.765 ± 0.01, 0.961 ± 0.004, and 0.838 ± 0.01 for 2011, 2012, and 2013, respectively).

Discussion

Although the establishment of non-seeded species can be controlled in restoration projects through the combination of seedbank exhaustion (Bakker & Poschlod 1996) and seed dispersal limitation (Bragg & Hulbert 2006), increased atmospheric N deposition has the potential to alter the success of future restoration efforts by disproportionately favoring the growth of adventive species that can establish and exploit high N availability (Miles & Knops 2009; Seabloom et al. 2013). However, contrary to this prediction, N addition did not significantly increase the cover or biomass of adventive species during 3 years in our study. Instead, native species that were seeded as part of the restoration increased in cover in response to N addition. This result was probably influenced strongly by limitations to the specific adventive species that were able to colonize the site. In particular, while the expectation that increased N availability will favor adventive species is often based on the relative abilities of native C_4 grasses versus adventive C_3 grasses to exploit the added N (Vinton & Goergen 2006; Doll et al. 2011), the latter species remained mostly absent from the site. Therefore, the preparation of the site through multiple glyphosate applications, coupled with the continued control of adventive grasses species at the periphery of the restoration site through glyphosate application, appeared to be successful in limiting the abundance of adventive species in the plots, despite the presence of added N.

The first few years after a disturbance can be critical for establishing the dominance of tallgrass species, because of the potential for rapid growth of pioneer species under the high light conditions (Wedin 2004). Although N addition did not facilitate the growth of adventive species, in the first year following our restoration N addition dramatically increased the abundance of *C. canadensis*, a native species whose seeds were not sown as part of the restoration. With the increase in total percent cover of all species in the second year, the abundance of *C. canadensis* declined sharply. This response was consistent with the colonization pattern of *C. canadensis*, which is a fast-growing, weedy species that is successful on high N soil, but is not able to compete effectively with tallgrass species for light over time (Tilman 1987; Thébaud et al. 1996; Prieur-Richard et al. 2000).

Despite the overall increase in native species cover with N addition, the native legumes that were part of the prairie restoration all declined. Declines in legume abundance with N addition are often observed in grasslands, given the reduced competitive advantage of legumes under high N availability (Skogen et al. 2011). However, contrary to the expectation that increased N addition decreases species diversity (Wedin & Tilman 1990; Reich et al. 2001; Clark & Tilman 2008; Bobbink et al. 2010), we did not observe a significant decrease in the Shannon diversity index. Such decreases might be expected to become more prevalent over time, as the plant canopy fills in and competition for light intensifies (Wedin 2004). Furthermore, as discussed above, the prevalence of bunch-forming grasses can permit the establishment and persistence of rarer species in gaps (McGregor et al. 1991). Several native C_3 grass species (*Bromus kalmii* and *Elymus trachycalus*) increased significantly in response to N addition in our plots, which is consistent with the expected response of C_3 grasses (Vinton & Goergen 2006; Doll et al. 2011). However, both of the latter C_3 species are bunchgrasses, and N addition effects on diversity may have been evident instead if the restoration plots had become dominated by C_3 turf-forming grasses.

The biological effects of chronic, low amounts of atmospheric N deposition can take decades to emerge in grasslands (Tilman 1990; Wedin & Tilman 1993; Stevens et al. 2010). Whereas N addition is typically one of the most influential factors in global change field experiments in grasslands (Dukes et al. 2005), such studies often apply very high rates of N that extend beyond those predicted for atmospheric N deposition in most regions (Bobbink et al. 2010). Nevertheless, at the functional group level, we generally observed similar responses to the addition rate of $2\,g\,m^{-2}\,yr^{-1}$, a moderate rate of N addition that is near the low estimate for deposition in our region in the coming decades, as we did for the addition rate of $6\,g\,m^{-2}\,yr^{-1}$, which exceeds the high estimate for the coming decades (Galloway et al. 2004). However, among the seeded, native, non-leguminous forbs, there were differences in species responses to moderate versus high N addition by the third year; a consortium of species responded positively to moderate N addition, whereas only one native forb species that was seeded (*Monarda fisulosa*) showed a strong response to high N addition. The lack of a positive response to high N by the consortium of forb species probably resulted from increased competition from *M. fistulosa*, and possibly other species, that flourished in the high-N plots.

Although we observed numerous plant cover responses to N addition, and percent cover is typically a good predictor of variation in biomass (MacDonald & Burke 2012), there were no significant N effects on aboveground biomass; total aboveground biomass increased in response to both levels of N addition, but the variability among samples was very high. The size of our biomass sampling quadrats (0.25×0.25 m) was consistent with that used elsewhere for prairie communities (Seastedt et al. 1991; Tilman & Wedin 1991; Biondini 2007; Dickson & Busby 2009; Socher et al. 2012; Seabloom et al. 2013), and was selected to minimize disturbance of the long-term plots. However, it was evident when sampling that the presence versus absence of a dominant species such as a bunchgrass in a given quadrat could influence the total biomass measure substantially, which explains why the variability among samples was high. Moreover, the biomass to cover ratio of the grasses was generally much higher than that of the forbs, which meant that the grasses (many of which did not increase significantly in cover in response to N addition) contributed proportionately higher to the total biomass estimate (relative to the cover estimates) than did the forbs. With the virtual absence of adventive C_3 grasses as competitors, the question arises as to why the native C_4 grasses did not exhibit a significant increases in biomass or cover in response to N addition. Although the mechanism remains speculative, this lack of a response could have resulted from competition from the native C_3 grasses and native forbs that increased in response to N, or possibly as a result of plant–microbial interactions that limited N availability to the C_4 grasses.

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